

Introduction to Causal Inference

The Potential Outcome Framework

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Causal Language Is Widely Used

- My headache went away because I took an aspirin
- She has a good job because she went to college
- She has long hair because she is female

Informal conclusions are drawn from association

- comparison between different individuals/ same individuals at different time points
- comparison of results of different actions/ manipulations

Association $\neq$ Causation

- We were reminded to **avoid making causal conclusions!!!**
- But what if causal statement is just what we seek???

Examples of Possible Causal Questions:

- What is the effect of a new drug on cholesterol reduction?
- What is the effect of offering a scholarship to a private grammar school on children's test scores?
- What is the impact of “blind” auditions on job offers for female musicians?
- Does he have lung cancer because he smokes?
- Does attending the job-training program increase his earnings?

We need to rigorously define and formulize **Causation!**

Treatment

- Treatment: An active intervention (the active treatment) applied at a particular moment in time, whose effects we wish to assess relative to no intervention or to another intervention (the control treatment).
- The active treatment and control treatment should be well defined.
- Treatment is an action, intervention, or manipulation.
- Causality is tied to a treatment applied to a unit.

No manipulation, no causation.

- Manipulation does not have to be performed, but should be theoretically possible – could be hypothetical
- e.g., the sun causes the planets to travel in their orbits
- Let's only consider settings with two actions.

Other Basic Concepts

- Unit: A physical object at a particular moment in time (you today and you tomorrow are distinct units).
- Covariate: A background characteristic of a unit that could not have been affected by treatment assignment (consider the time).
- Potential Outcomes: Outcomes that would be observed if active treatment is applied and that would be observed if control treatment is applied. (Also called Counterfactual Outcomes.)

Individual Causal Effect

- Comparison of potential outcomes under active treatment and control treatment, **for each unit**.
- Treatment (two possible values 0 and 1) has a causal effect on a unit's outcome Y if
$$Y(0) \neq Y(1).$$
- Here $Y(0)$ is the outcome under control and $Y(1)$ is the outcome under (active) treatment.
- Measures for individual causal effect: $Y(1) - Y(0)$; $Y(1)/Y(0)$

Hypothetical Example With One Unit

The unit is you at a particular point in time with a headache.

Covariate (X) is your assessment of your initial headache before taking the treatment.

Outcome (Y) is your assessment of your headache pain two hours after taking an aspirin (treatment) or not taking aspirin (control).

Unit	Covariate	Potential Outcomes		Causal Effect of treatment on Y
	X	$Y(1)$	$Y(0)$	$Y(1) - Y(0)$
You	90	20	60	$20 - 60 = -40$

Note: $Y - X$ is not the causal effect.

Hypothetical Example With One Unit —adjusting with X

The unit is you at a particular point in time with a headache.

Covariate (X) is your assessment of your initial headache before taking the treatment.

Outcome (Y) is your assessment of your headache pain two hours after taking an aspirin (treatment) or not taking aspirin (control).

Unit	Covariate	Potential Outcomes		Causal Effect of treatment on Y
	X	$Y(1)$	$Y(0)$	$(Y(1) - X) - (Y(0) - X)$
You	90	20	60	$(20 - 90) - (60 - 90) = -40$

The Fundamental Problem of Causal Inference

- **A missing data problem**: at most one of the potential outcomes can be observed!

Some notes:

- The individual causal effect is the comparison of potential outcomes, for the same unit, at the same moment in time.
- The definition of the causal effect does not depend on which outcome is actually observed. That is, it does not depend on the assignment of treatment.
- The potential outcomes do not change if the treatment assignment switches, but the observed outcome does! (rigorous formulation provided later)
- **Individual causal effect is unidentifiable because of the missing potential outcomes.**

Average Treatment Effect

- ATE:

$$\tau_{ATE} = E(Y_i(1) - Y_i(0)) = E(Y_i(1)) - E(Y_i(0))$$

may also be called ACE (average causal effect)

- Often, we focus on estimating the average treatment effect in the sample: Sample Average Treatment Effect (SATE)

$$\tau_{fs} = \frac{1}{N} \sum_{i=1}^N (Y_i(1) - Y_i(0))$$

- Variation from treatment assignment (the missing potential outcome)
- This is like assuming the finite sample is the population

- Compare to the Population Average Treatment Effect (PATE)

$$\tau_{sp} = E_{sp}(Y_i(1) - Y_i(0))$$

- Variation from treatment assignment and sampling process
- Requires knowledge about the sampling process (like the common statistical inference)

SATE & PATE

- If your sampling was random from the super population, then

$$E(\text{SATE}) = \text{PATE}$$

For this reason, we are often sloppy in distinguishing between the two quantities.

Other Average Effects

- Average treatment effect for the treated

$$\begin{aligned}\tau_{ATT} &= E(Y_i(1) - Y_i(0) \mid W_i = 1) \\ &= E(Y_i(1) \mid W_i = 1) - E(Y_i(0) \mid W_i = 1)\end{aligned}$$

- For random treatment assignment,

$$Y_i(0) \perp W_i, Y_i(1) \perp W_i$$

$$\begin{aligned}E(Y_i(0) \mid W_i = 1) &= E(Y_i(0) \mid W_i = 0) = E(Y_i(0)) \\ E(Y_i(1) \mid W_i = 1) &= E(Y_i(1) \mid W_i = 0) = E(Y_i(1))\end{aligned}$$

Then ATT=ATE

- Average treatment effect for the untreated

$$\begin{aligned}\tau_{ATU} &= E(Y_i(1) - Y_i(0) \mid W_i = 0) \\ &= E(Y_i(1) \mid W_i = 0) - E(Y_i(0) \mid W_i = 0)\end{aligned}$$

Other Effect Measures (Estimands)

- ▶ $E[Y(1)] - E[Y(0)]$ (mean difference)
- ▶ $Med[Y(1)] - Med[Y(0)]$ (median difference)
- ▶ $E[Y(1)]/E[Y(0)]$ (risk ratio)
- ▶ $\frac{Pr[Y(1)=1]/Pr[Y(1)=0]}{Pr[Y(0)=1]/Pr[Y(0)=0]}$ (odds ratio, when Y is binary)

None of these can be calculated or observed, but they can be defined and estimated

Fundamental Requirements and Assumptions

How do we learn about the missing potential outcomes and estimate causal effects?

We need multiple units:

- At least one unit receives active treatment and at least one unit receives control treatment.
- Potential outcomes $Y_i(0)$ and $Y_i(1)$ are well-defined functions of the unit label and the treatment indicator
- SUTVA (Stable Unit Treatment Value Assumptions)
 - No interference among units, i.e., the potential outcomes for any unit do not vary with the treatments assigned to other units.
 - No hidden variations of treatments, i.e., there are no different forms or versions of treatment levels
- Causal inference is generally impossible without such assumptions!

More on SUTVA

- Each version of treatment corresponds to a different potential outcome.
- SUTVA can depend on definition of treatment
 - (Any random aspirin) vs (no aspirin): coarse
 - (Aspirin 500 mg) vs (no aspirin): refined
- SUTVA can depend on the choice of outcome
 - Y = mild versus severe: coarse
 - Y = actual intensity from 0 to 100: refined
- We are studying different causal effects under different definitions of treatment and choices of outcome.
- The point: SUTVA implies that the potential outcomes are well defined for each unit and each treatment

An Estimate of ATE

Define the **treatment assignment** variable W_i

$$W_i = \begin{cases} 0, & \text{if the } i\text{th unit is assigned to control;} \\ 1, & \text{if the } i\text{th unit is assigned to treatment.} \end{cases}$$

$$Y_i^{obs} = Y_i(W_i) = \begin{cases} Y_i(0), & \text{if } W_i = 0; \\ Y_i(1), & \text{if } W_i = 1. \end{cases}$$

$$Y_i^{mis} = Y_i(1 - W_i) = \begin{cases} Y_i(1), & \text{if } W_i = 0; \\ Y_i(0), & \text{if } W_i = 1. \end{cases}$$

$$\bar{Y}_c^{obs} = \frac{1}{N_c} \sum_{i: W_i=0} Y_i^{obs}, \quad \bar{Y}_t^{obs} = \frac{1}{N_t} \sum_{i: W_i=1} Y_i^{obs}$$

An intuitive estimate of ATE (SATE or PATE) is

$$\hat{\tau} = \bar{Y}_t^{obs} - \bar{Y}_c^{obs}$$

Sources of Errors for Estimations

Two main sources of errors for the estimations of causal effects:

- Random counterfactuals: error brought by W

Error for estimating SATE or PATE

Compare $\hat{\tau} = \bar{Y}_t^{obs} - \bar{Y}_c^{obs}$ and $\tau_{fs} = \frac{1}{N} \sum_{i=1}^N (Y_i(1) - Y_i(0))$

- Sampling Variability:

Error for estimating PATE

$Y_i(0), Y_i(1), i = 1, \dots, N$ are samples from the super population

Compare $\tau_{sp} = E_{sp}(Y_i(1) - Y_i(0))$ and $\tau_{fs} = \frac{1}{N} \sum_{i=1}^N (Y_i(1) - Y_i(0))$

Introduction to Assignment Mechanism

- W : heart transplant surgery assignment (1: treated, 0 control)
- Y : five-day survival (1: death, 0 survival), X : pre-condition covariate (1: critical, 0: noncritical)
- Let's suppose we have 80 units with each one here representing 10 units (to mimic big data)

Unit	Treatment Assignment	Potential Outcomes	
	W	$Y(1)$	$Y(0)$
1	1	1	?
2	0	?	1
3	1	1	?
4	1	0	?
5	0	?	0
6	1	0	?
7	0	?	1
8	0	?	0

Unit	Covariate	Treatment Assignment	Potential Outcomes	
	X	W	$Y(1)$	$Y(0)$
1	1	1	1	?
2	1	0	?	1
3	1	1	1	?
4	1	1	0	?
5	0	0	?	0
6	0	1	0	?
7	0	0	?	1
8	0	0	?	0

$$\hat{\tau} = \bar{Y}_t^{obs} - \bar{Y}_c^{obs} = \frac{2}{4} - \frac{2}{4} = 0$$

No causal effect!

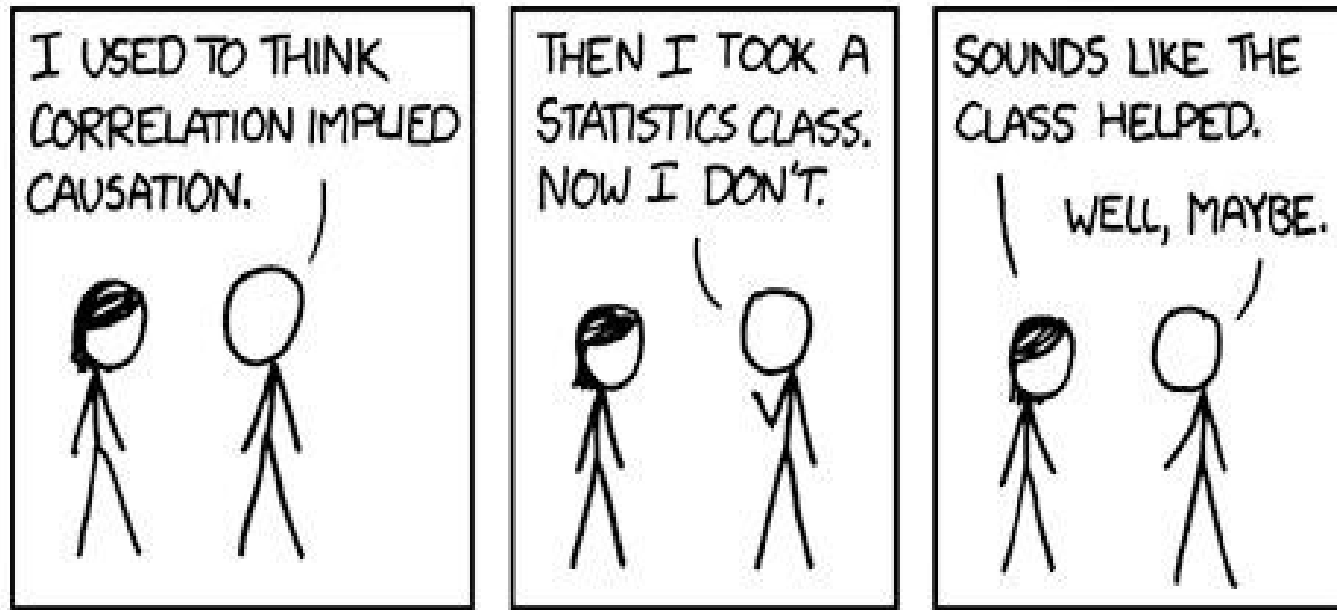
$$\hat{\tau}(x=1) = \bar{Y}_t^{obs}(x=1) - \bar{Y}_c^{obs}(x=1) = \frac{2}{3} - 1 = -\frac{1}{3}$$

Significant causal effect!

$$\hat{\tau}(x=0) = \bar{Y}_t^{obs}(x=0) - \bar{Y}_c^{obs}(x=0) = 0 - \frac{1}{3} = -\frac{1}{3}$$

Assignment Mechanism

- Why contradictory results? Which one to believe?
- Depending on the treatment assignment mechanism!
- Assignment mechanism is the process governing which units receive treatment and which receive control.
- Information of treatment assignment mechanism is significant and may affect causal conclusions!



Questions?

Let us know something about you!